

House Price Prediction Using Linear Regression

An Analysis of the California Housing Dataset

Abstract

This report details the development of a Linear Regression model aimed at predicting house prices leveraging the widely utilized California Housing Dataset. The end-to-end workflow encapsulated in this study includes data ingestion, rigorous exploratory data analysis (EDA), data preprocessing, model training, and comprehensive evaluation. The performance of the predictive model is assessed using standard regression metrics, namely Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the R^2 Score, demonstrating the practical and effective application of foundational machine learning algorithms to real-world regression challenges in the real estate domain.

1. Introduction

House price prediction is a canonical problem within the field of machine learning, serving as a vital tool for various stakeholders in the real estate market. Accurate pricing estimates empower home buyers, sellers, and corporate investors to make data-driven, informed financial decisions. This project utilizes **Linear Regression** as a robust, baseline supervised learning algorithm to estimate property values based on a multiplicity of housing-related features.

2. Dataset Description

The analysis is grounded in the **California Housing Dataset**, which contains aggregated housing metrics collected during the California census. The dataset comprises several predictive features and one continuous target variable:

Feature Name	Description
Median Income	Median income of households in the block group.
House Age	Median age of the houses within the block group.

Feature Name	Description
Average Rooms	Average number of rooms per household.
Average Bedrooms	Average number of bedrooms per household.
Population	Total population residing in the block group.
Occupancy	Average number of household members.
Latitude / Longitude	Geographical coordinates of the block group.
Median House Value	Target Variable: The median value of homes in the block group.

3. Exploratory Data Analysis (EDA)

A comprehensive Exploratory Data Analysis (EDA) was executed to unearth underlying structures and relationships within the dataset. Initial steps involved computing summary statistics to gauge the central tendencies and dispersions of the features . Subsequent correlation analyses and distribution plots were pivotal in identifying feature-target relationships. Notably, the **Median Income** exhibited the most substantial positive correlation with the target variable, indicating a strong relationship with house prices .

4. Methodology

The machine learning pipeline was structured to ensure unbiased model evaluation. The dataset was partitioned into a **training set (80%)** and a **testing set (20%)**. A Linear Regression model was instantiated and trained utilizing the Scikit-Learn library. During the training phase, the algorithm calculated the optimal coefficients (weights) for each feature, establishing the mathematical relationship between the housing attributes and the corresponding house prices.

5. Model Evaluation

To quantify the predictive efficacy of the linear model, predictions were generated and evaluated. The model's performance was measured using the following core metrics:

- **Mean Absolute Error (MAE):** Measures the average magnitude of the prediction errors.
- **Root Mean Squared Error (RMSE):** Evaluates the standard deviation of the prediction errors.
- **R² Score (Coefficient of Determination):** Assesses how well the regression predictions approximate the real data points.

These metrics holistically measure prediction accuracy and model effectiveness. Visual diagnostic tools, such as an *Actual vs. Predicted* scatter plot, can be further employed to visualize model performance and confirm reliability.

6. Conclusion and Future Work

This project successfully illustrates an end-to-end machine learning workflow applying Linear Regression to a complex dataset. The current model establishes a strong, interpretable baseline for house price prediction.

Future trajectories for improvement include :

- **Feature Engineering:** Creating or deriving new features to improve accuracy.
 - **Advanced Algorithms:** Exploring models such as Polynomial Regression, Random Forests, and Gradient Boosting techniques.
-